

Cracks and root channels promote both static and dynamic vertical hydrological connectivity in the Yellow River Delta

Yuan Cui^a, Ying-Hu Zhang^b, Shi-Jun Zhou^a, Yue-Yan Pan^a, Rui-Qing Wang^a, Zhen Li^a, Zhen-Ming Zhang^{a,*}, Ming-Xiang Zhang^{a,**}

^a School of Ecology and Nature Conservation, Beijing Forestry University, Beijing, 100083, China

^b School of Forestry, Nanjing Forestry University, Nanjing, 210037, China

ARTICLE INFO

Handling Editor: M.T. Moreira

Keywords:

Coastal wetlands
Hydrological connectivity
Infiltration rate
Solute transport

ABSTRACT

Vertical hydrological connectivity generally related heterogeneity of soil water and nutrient and the sustainability of vegetation restoration. Both the static and dynamic connectivity of soil have been studied for years, the relationship between them remains unclear, especially for coastal wetlands that greatly influenced by hydrological processes. In this study, static connectivity was defined as water flow paths and quantified by dye tracer experiment, whereas dynamic connectivity refers to the process of water movement and solute transport and described by solute penetration. Relationships between them were analyzed through the nonlinear fitting. Results showed that: 1) static connectivity was construed mainly by cracks in the upper soil layer and by root channels in the deeper soil, and these macropores led to heterogeneous distribution of water and solutes; 2) dynamic connectivity differed between soil columns, and non-equilibrium flow characterized by high infiltration rate and concentration in the initial phase was observed in soil with cracks and root channels; and 3) the static connectivity had significant effects on water movement while had no effects on solute transport. Infiltration rate decreased as dye coverage, fractal dimension, and the number of water pathway logarithmically increases. These findings highlight the importance of cracks and root channels on vertical hydrological connectivity in soil of *S. salsa* community, and suggest linking the static structure of water flow paths with soil water dynamic and heterogeneity to understand soil functions.

1. Introduction

Coastal wetlands provide a wide range of ecosystem services (valued up to US \$193,845 ha⁻¹ yr⁻¹), including the support of commercial fisheries, carbon sequestration, and water purification (Barbier et al., 2011; Brown et al., 2021; Costanza et al., 2014). However, 16.02% of tidal flats and at least 25% of salt marshes were lost due to human activities as well as climate change in the past 30 years (McOwen et al., 2017; Murray et al., 2019). Restoration has been elevated as a global strategy to enhance the function and services of coastal ecosystems (Bayraktarov et al., 2016; Liu et al., 2016, 2021). Soil hydrology is one of the critical limiting factors that affects vegetation survival, growth and restoration (Liu et al., 2020, 2021; Wu et al., 2016). Decrease in soil water could endanger the sustainability of vegetation restoration (Jia et al., 2019). Despite a considerable amount of money has been invested in coastal restoration, some projects either failed or yielded limited

results due to the neglect of hydrology at the watershed scale, e.g. river discharge from rivers to the ocean (Liu et al., 2021).

Hydrological connectivity provides a systematic and comprehensive perspective at different scale for explaining ecological process concerned water movement, like nutrients circulation, animal migration, and life cycle of the plant (Fernández et al., 2020; Hiatt and Passalacqua, 2015; Liu et al., 2019a, 2020; Yin et al., 2020). It refers to the transport of matter, energy, and organisms driven by water within or between elements of the hydrologic cycle (Freeman et al., 2007; Pringle, 2001). A better understanding on hydrological connectivity of coastal wetland will support coastal management and conservation theoretically and practically, as well as make large-scale restoration programs more cost efficient.

Hydrological connectivity is 3-dimensional: longitudinal, lateral, and vertical (Covino, 2017). Longitudinal connectivity is the most studied within a relatively large scale, and it refers to the water

* Corresponding author.

** Corresponding author.

E-mail addresses: zhenmingzhang@bjfu.edu.cn (Z.-M. Zhang), zhangmingxiang@bjfu.edu.cn (M.-X. Zhang).

<https://doi.org/10.1016/j.jclepro.2022.132972>

Received 28 October 2021; Received in revised form 18 May 2022; Accepted 30 June 2022

Available online 8 July 2022

0959-6526/© 2022 Elsevier Ltd. All rights reserved.

movement along with the tidal channel or river courses (Granados et al., 2020; Wohl and Beckman, 2014). The rate of flow, water depth, current velocity and other parameters change as time are usually used to quantify the dynamic connectivity (Abdullah and Jain, 2020), while the topological structure of the river or channel network are the key parameters for the static connectivity (Bishop-Taylor et al., 2018; Cui et al., 2020; Liu et al., 2019a; Tan et al., 2021). Lateral connectivity refers to the water movement on the orthogonal direction to channel (Covino, 2017). The dynamic connectivity is basically measured by surface or subsurface flow while the static connectivity is usually quantified by topography, preferential pathway, or the soil hydrological characteristics (Biehler et al., 2020; Ward, 1989).

Recently, hydrological connectivity inside soils is gradually being recognized (Dai et al., 2020; Zhang et al., 2021b), as it was generally related to plant growth, soil nutrient redistribution, surface runoff, and groundwater replenishment and pollution (Franklin et al., 2021; Guo et al., 2019; Wu et al., 2021; Zhang et al., 2021b). Vertical hydrological connectivity evaluates the static structure of flow pathway and dynamic process of water movement and solute transport from the soil surface to the deep layer (Zhang et al., 2021a). As static hydrological connectivity refers to the pattern of geological units which influence the processes of water-mediated transport, and for longitudinal, usually in a large scale, static hydrological connectivity is mainly parameterized by the structure of flow pathway (topology of river/channel network) (Bishop-Taylor et al., 2018; Tan et al., 2021). We thus define static vertical hydrological connectivity as the water flow pathway in soil layers. Soil macropores, such as root channels, cracks/fissures, and soil fauna channels usually act as water flow pathways in soils (Bracken et al., 2013). Dye tracer experiments are widely used to visualize structure of water flow pathway (Morris et al., 2008), which are semi-quantitatively evaluated based on stained areas of vertical and horizontal soil profiles (Dai et al., 2020).

Dynamic hydrological connectivity tends to describe how the spatial-temporal distribution of water channels enable matter transportation, and puts more emphasis on the driving force and the process (Zhang et al., 2021a). Therefore, vertical dynamic hydrological connectivity was defined as the process of water movement from one point to another in soil profiles. For coastal wetland, infiltration as well as water-mediated transport of matters are the major vertical hydrological processes and thus it can be assessed by solute penetration, that is, variations in the infiltration rates and concentrations with time (Jiang et al., 2019).

Static hydrological connectivity may have a significant influence on the dynamic transport processes. While the relationship between static and dynamic vertical hydrological connectivity remains unclear. Usually, macropore networks comprised of cracks, roots channels, and earthworm burrows can promote water movement and solute transport (Katuwal et al., 2015; Wang et al., 2021). In a soil profile, the presence of a dense network of macropores increased the infiltration rates in hydrophobic soils (Burch et al., 1989). However, this does not mean that the infiltration rate or solution concentration always changes linearly with the structure of flow paths. Water movement and solute transport are also influenced by multiple factors such as pore binding force and pore water velocity. Quantitative analysis of the relationship between the static and dynamic connectivity will help to reveal the effects of wetlands on plant growth and other hydrology function, such as runoff, and groundwater replenishment and pollution.

Existed researches concerned infiltration in land use types closer related to production or other human activities such as in agroforestry systems (Guo et al., 2019; Jiang et al., 2019) and forests (Zhang et al., 2019). While, there remains a gap about coastal wetlands, despite its valuable ecological functions in habitat support and microclimate regulation (Browner et al., 2021). A better understanding on hydrological cycle in coastal wetland will support coast management and conservation theoretically and practically, especially considering the increasing frequency and intensity of extreme precipitation events in the context of

global climate change (Du et al., 2019; Wang et al., 2013a). Since 2000, the Chinese government funded 36 projects aimed at protecting current natural wetlands and restoring damaged wetlands (Liu et al., 2019a). The Yellow River Delta (YRD) is one of the priority areas due to its sensitivity and importance ecologically and economically.

In this study, static and dynamic vertical hydrological connectivity were studied simultaneously. We defined static vertical hydrological connectivity as the water flow paths in soil layers. Besides, dynamic vertical hydrological connectivity was simulated by monitoring the movement of hydraulic dye tracer through soil columns. The aims of this study were three-fold: (1) parameterization the static hydrological connectivity by quantify the morphological structure of flow path (macropore) networks, (2) quantifying dynamic hydrological connectivity by infiltration rate and relative concentration of the effluent with time for the soil columns, and (3) link the static connectivity to dynamic connectivity to reveal the influence of hydrological connectivity structure on its process in the YRD within the dominant plant communities.

2. Materials and methods

2.1. Study site

We conducted our field study in the Yellow River Delta (37°49'27"N, 119°3'36"E), which is in the southwest of the Bohai. The study area is characterized by warm temperate monsoon climate with an annual precipitation of 551.6 mm and an annual average evaporation of 1928.2 mm (Feng et al., 2021). Groundwater depth in study area is shallow, with an average of ~1.0–2.5 m (Xia et al., 2020). The proportion of silt, sand, and clay in the soils of the Yellow River Delta is 54.2%, 37.03%, and 8.77%, respectively, and fluvo-aquic is the most widely distributed soil type (Li et al., 2019; Liu et al., 2019b). The dominant vegetation includes *Suaeda salsa*, *Tamarix chinensis*, and *Phragmites australis*.

2.2. Experiment design

We aim to parameterize both static and dynamic vertical hydrological connectivity. Three parallel plots (SP1, SP2, and SP3) with *S. salsa* (0.2 m in height) were randomly selected at intervals of 100 m, and they have similar vegetation coverage (80%), soil types, and terrain characteristics.

Dye tracer experiments were conducted in September 2017 to mark the water flow pathway under *S. salsa* communities in the YRD. Vegetation was cleared off before sampling. Brilliant Blue FCF (For Coloring Food; C.I.42090) dye solution (4 g L⁻¹, total 50 L) was uniformly sprayed across a 1.2 × 1.2 m area in the center of each experimental site (Fig. 1). Approximately 24 h after dye tracing experiment, the central part of each plot (1.0 × 1.0 m, to avoid boundary effects) was selected for the excavation of the vertical and horizontal profiles. Three parallel vertical soil profiles were excavated, starting from the outside of experimental plot to the center of the experiment plot. Subsequently, nine horizontal profiles at depths of 1, 5, 10, 15, 20, 30, 40, 50, and 60 cm were excavated from the remaining soil body. Soil profiles were then recorded with a digital camera.

Pictures were transformed into binary images in black and white following the correction of geometric distortion and adjustment of hue, saturation, brightness, and other parameters in Photoshop CC 2017. The depth from the soil surface to the maximum depth of dye infiltration was used for the analysis. We extracted the stained areas from these binary images using Tabulated Area tool in ArcGIS 10.2 (ESRI Inc., Redlands, California, U.S.). This tool calculates the stained area within different zones in the image. Images (1000 × 600 mm) were divided into 100 strip-like polygons (1000 mm in width and 6 mm in height). These output data were then used to calculate indices describing water flow pathway.

Solute penetration experiments were conducted in December 2017 in the laboratory to simulate variations in the infiltration rates and

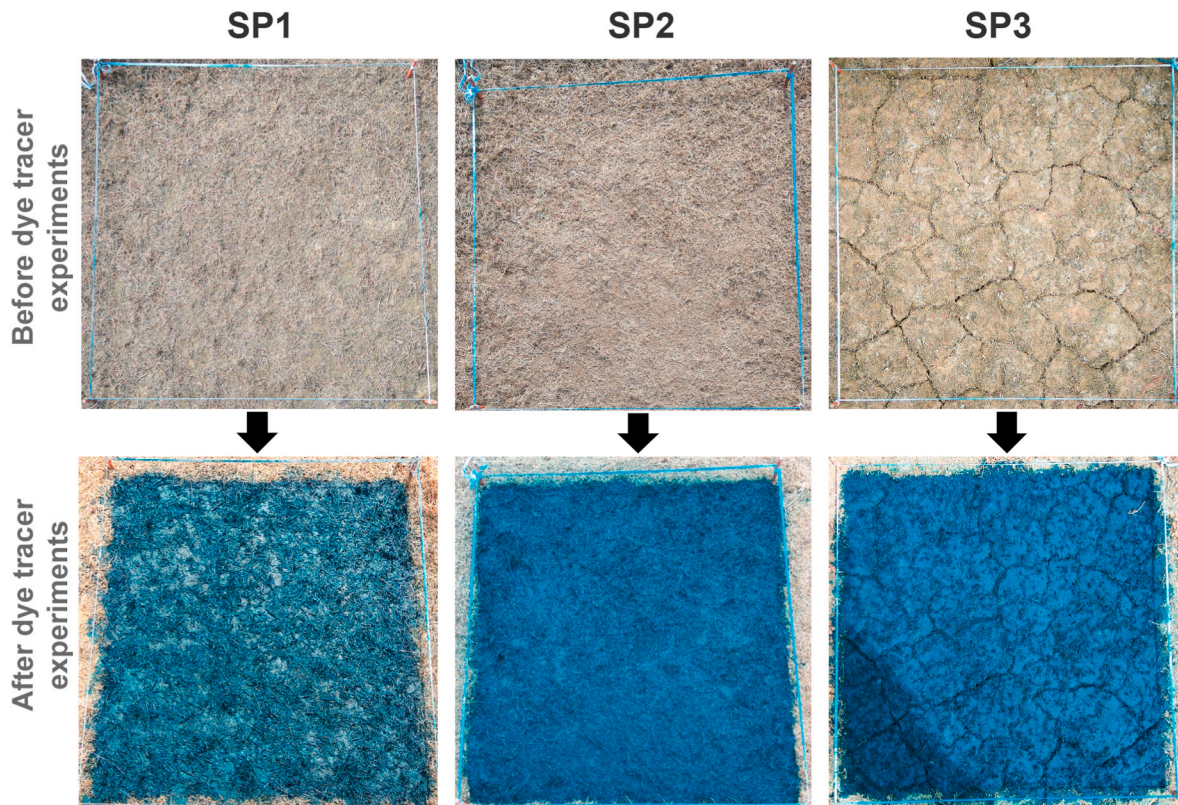


Fig. 1. Dye tracer experiments at three experimental plots in the field. SP1, SP2, and SP3 refer to the three parallel plots with *Suaeda salsa*.

concentrations with time. 22 well-preserved soil cores with a diameter of 7 cm and height of 10 cm were collected at the depths of 0–10 cm (column 1–5), 10–20 cm (column 6–10), 20–40 cm (column 11–16), and 40–60 cm (column 17–22). Brilliant Blue FCF dye solution (1 g L^{-1}) was applied to the top of the soil column, with a constant hydraulic head of 1 cm. After the colored solution permeated through soil columns, the effluent was collected using polyethylene flasks every 1, 2, 5, and 10 min until the flow rate stabilized.

Besides that, we also collected soil samples from the same three plots at depths of 0–10, 10–20, 20–40, and 40–60 cm and measured physicochemical properties, including soil water content by mass (wt. %), soil bulk density (BD, g cm^{-3}), total soil porosity (TP, %), and soil organic carbon (SOC, g kg^{-1}). Soil water content was the proportion of mass reduced after oven-drying at 105°C to constant mass of the soil at the end of the drying. Organic carbon content was determined using dichromate oxidation (Nelson and Sommers, 1983). Soils collected by a stainless-steel cutting ring were measured for BD. Total soil porosity was calculated as the following equation:

$$TP = (1 - BD/ds) \times 100$$

where ds refers to the soil particle density (g cm^{-3}) and was considered to be 2.65 g cm^{-3} (Jiang et al., 2019).

2.3. Parametrization of static vertical hydrological connectivity

Seven parameters based on the dye experiments were selected to quantify the static connectivity: (1) dye coverage (DC, %), the percentage of the stained area to the total area of soil profile (Flury et al., 1994b); (2) uniform infiltration depth, which represents the depth of the uniform infiltration front and can be identified when dye coverage decreases to 80% (van Schaik, 2009); (3) maximum depth of dye infiltration: defined as the maximum depth that dye tracers can reach (Zhang et al., 2015a); (4) the number of water pathways: extracted by Image-Pro Plus software based on binary images of the soil profile (Dai

et al., 2020); (5) fractal dimension (FD): represents the complexity of the water pathways of soil profile. $FD = \lim_{\varepsilon \rightarrow 0} [\log N(\varepsilon) / \log(1/\varepsilon)]$, where ε is the average width of each stained area and $N(\varepsilon)$ is the number of fractals covered by this dye-stained area (Ogawa et al., 1999; Zhang et al., 2021b). Generally, the larger the fractal dimension, the more irregular the distribution of water pathways and the greater the spatial heterogeneity (Dai et al., 2020); (6) length index (LI): sum of the absolute differences of dye coverage between two adjacent soil layers in vertical profile, $LI = \sum_{i=1}^{100} |DC_{i+1} - DC_i|$, where i represents a given depth interval (or zone) of the 100 in which dye coverage (DC, %) was calculated (Bargués Tobella et al., 2014); (7) peak index: number of times that the vertical line defined by the total dye coverage intersects the dye coverage profile (Bargués Tobella et al., 2014). The last three parameters are used for describing the heterogeneity of water pathways (Benegas et al., 2014; Dai et al., 2020).

2.4. Quantifying dynamic vertical hydrological connectivity

The dynamic vertical hydrological connectivity was quantified by infiltration rate and relative concentration of solution. The infiltration rate (mm h^{-1}) was defined by the depth of effluent per unit time per unit area. The relative concentration was normalized by dividing the measured concentration (C) by the applied concentration (C_0 , 1 g L^{-1}), consequently, the measured concentration (C) was exactly we need and changed between 0 and 1 (Zhang et al., 2015a). The concentration of dye in the effluent was measured with an ultraviolet (UV) spectrophotometer at a wavelength of 627 nm. Relationships between static hydrological connectivity and dynamic hydrological connectivity were analyzed through the nonlinear fitting in R 4.0.5.

3. Results and discussion

3.1. Initial soil properties

Average initial soil water content, BD, TP, and SOC were $23.41 \pm 1.26\%$, $1.44 \pm 0.08 \text{ g cm}^{-3}$, $46.14 \pm 2.41\%$, and $2.06 \pm 0.94 \text{ g kg}^{-1}$, respectively. There were no differences in the physical properties between any two soil layers, while significant higher SOC in 0–10 cm soil compared to other soil layers (Table 1). The *S. salsa*'s root system mainly distributes in 0–10 cm surface soil (Chen and Sun, 2020; Chen et al., 2016), and thus more rhizodeposits from living roots were input into this layer. Moreover, plant litters after plant death were also contributes greatly to SOC in surface soil.

The significance level is 0.05. Significant differences between soil depths are represented by different lower-case.

3.2. Static hydrological connectivity

Static hydrological connectivity refers to the morphological structure of macropore network, which was quantified by dye coverage, uniform infiltration depth, maximum depth of dye infiltration, the number of water pathways, fractal dimension, length index, and peak index. The results are shown in Table 2 and the visualization are shown in Fig. 2 and Fig. 3.

A large fraction of dye-tracers being channelized via the complex network composed of cracks and root channels (Figs. 2 and 3). These irregular branches of cracks and root channels meant water and solute will bypass soil matrix (Alaoui et al., 2011) and migrate directly to the deep soil layers (Benegas et al., 2014; Qi et al., 2020; Sheng et al., 2014; Zhang et al., 2015b). Our study observed a near-surface uniform infiltration depth (0.4 cm) and a deeper maximum dye infiltration depth (about 105 cm, Table 2). This was consistence with previous findings that maximum infiltration depth was $\sim 96 \text{ cm}$ in alfalfa grassland (Guo et al., 2019). Macropores provided by cracks and root channels usually reduce the time for water to percolate to deep layers and also reduce evaporation losses from the topsoil layers, which together will lead to enhanced deep drainage and potentially groundwater recharge (Bargués Tobella et al., 2014; Bordoloi et al., 2020). Considering the petroleum exploitation and agricultural reclamation in this area (Wang et al., 2017), the presence of cracks and deep roots may lead to the loss of applied resources and energy and also an increase of pollution risk in groundwater.

Cracks are common under field drying conditions and its occurrence and spatial patterns depend on soil properties (Franklin et al., 2021; Xu et al., 2022). Compared with peatland (SOC, 51.3 g kg^{-1} ; BD, 0.077 g cm^{-3}) (Carless et al., 2021), the SOC was low ($1.37\text{--}3.40 \text{ g kg}^{-1}$) and the BD was high ($1.36\text{--}1.53 \text{ g cm}^{-3}$) in this study. Lower SOC and higher BD often paired with soil compaction which will lead to soil cracking under drought condition (Jarvis, 2020; Nimmo, 2021). Moreover, soils in this region were dry as a results of high evaporation (Aridity index, 3.56) and restricted water supplement from tide water owing to the cofferdam (Liu et al., 2019b; Wang et al., 2013b). These conditions favor the formation

and propagation of desiccation cracks which then creates a soil environment conducive to water movement. Cracks can propagate in deeper soil after multiple cycles of drying-wetting (Bordoloi et al., 2020) and often exhibit temporal stability, i.e. flow repeatedly takes place through the same flow paths (Franklin et al., 2021).

Besides, channels made by living root as well as decayed roots also act as water flow pathways and can be responsible for enhancing water movement and solute transport (Beven and Germann, 2013; Zhang et al., 2021b). Generally, the *S. salsa*'s root system mainly distributes in 0–10 cm surface soil and is characterized by short taproot, dense lateral roots, and a small-diameter (Chen and Sun, 2020; Chen et al., 2016). Still, the distribution of roots of *S. salsa* in the deep soil was also observed in this study (Fig. 2). This might be the positive response of *S. salsa* to the dry environment and can help them with the acquisition of water and nutrients. Roots channels can connect with the pre-existing pores to improve continuity of water flow path and provide a temporary channel for macropore flow when *S. salsa* dies and decays (Guo et al., 2014; Zhang et al., 2017, 2021b). The water flow paths provided by cracks and root channels led to heterogeneous distribution of water and solutes in the soil.

Another finding through the dye tracer experiment was the hydrological barrier at the depth about 25–40 cm, which contribute to horizontal hydrological connectivity rather than vertical hydrological connectivity (Chelli et al., 2018; Zhang et al., 2017). It was supported by variation in dye coverage (Figs. 2 and 3). Mean dye coverage of vertical soil profile non-linearly decreased with increasing soil depth, presenting an increase around hydrological barrier layers (about 30, 25, and 35 cm for SP1, SP2, and SP3, respectively) (Fig. 2). A large amount of sediment carried by the Yellow River deposited around the river delta under complex hydrodynamic conditions and then formed different soil layers (Li et al., 2018). These layers may have a contrasting texture, structure, or elemental composition (Bracken et al., 2013; Li et al., 2019), and those with higher BD and viscosity may form the hydrological barrier (Chelli et al., 2018).

Almost all the cracks distributed above the barrier while beneath the barrier, barely cracks and only root channels can be found (Figs. 2 and 3). Thus, in the surface layer, static hydrological connectivity mainly construed by cracks while in the deeper layer, root channel is the main flow path. Evaporation losses from the topsoil layers led to soil cracks, and the existence of hydrological barrier protected deep soil from further cracking. Moreover, difference in bioturbation might lead to the heterogeneity in water flow paths above and below the hydrological barrier (Chelli et al., 2018; Zhang et al., 2017). Crab species, such as *Helice tridens*, *Macrophthalmus dilatatum*, live and waterfowl species, such as *Grus japonensis*, forage mainly in surface layer (Xiao et al., 2019; Yu et al., 2021). Those activities constructed the cracks while in the deeper layer, only root growth can create macropores. The deep root pathways allow plant to acquire water from deep soil layers under dry environments (Bargués Tobella et al., 2014; Stone and Kalisz, 1991).

Water flow paths mainly composed of cracks and root channels in soil of *S. salsa* community had a relatively unimpeded condition for vertical connectivity structure. This was showed by high values of the number of water pathways (652), fractal dimension (375), length (2.44), and peak indices (9.11) (Table 2). Fractal dimension, length indices, and peak indices give the information of the heterogeneity of morphological features of flow pathways (Bargués Tobella et al., 2014). These three indices were higher in our study areas than those of areas with Chinese tamarisk in the YRD (Dai et al., 2020), indicating that flow pathways were more irregular and complexity in soil of *S. salsa* communities.

3.3. Dynamic hydrological connectivity

The dynamic hydrological connectivity was quantified by both infiltration rate and solute concentration changes (Fig. 4). Results showed that both the initial infiltration rate and concentration were high in soils with more macropores, here the cracks and root channels (e.

Table 1
Soil physical and chemical properties at different soil depths (Mean $\pm$ S.D.).

Soil depth (cm)	Initial water content by mass (wt. %)	Bulk density (g cm^{-3})	Total soil porosity (%)	Soil organic carbon (g kg^{-1})
0–10	$24.45 \pm 0.98\text{a}$	$1.36 \pm 0.09\text{a}$	$48.92 \pm 2.61\text{a}$	$3.40 \pm 0.87\text{a}$
10–20	$23.39 \pm 0.32\text{a}$	$1.44 \pm 0.00\text{a}$	$45.47 \pm 0.46\text{a}$	$1.45 \pm 0.27\text{b}$
20–40	$23.77 \pm 0.86\text{a}$	$1.42 \pm 0.05\text{a}$	$45.99 \pm 0.65\text{a}$	$2.01 \pm 0.38\text{b}$
40–60	$22.04 \pm 1.75\text{a}$	$1.53 \pm 0.05\text{a}$	$44.17 \pm 3.05\text{a}$	$1.37 \pm 0.93\text{b}$

Table 2Static hydrological connectivity parameters of different sites (Mean $\pm$ S.D.).

Sites	Dye coverage	Uniform infiltration depth (cm)	Maximum dye infiltration depth (cm)	The number of water pathways	Fractal dimension	Length index	Peak index
SP1	0.12 $\pm$ 0.02a	0.40 $\pm$ 0.01a	89.7 $\pm$ 24.5b	503 $\pm$ 161a	263 $\pm$ 94a	2.29 $\pm$ 0.50a	9.00 $\pm$ 3.29a
SP2	0.10 $\pm$ 0.05a	0.50 $\pm$ 0.01a	132.2 $\pm$ 11.0a	713 $\pm$ 241a	375 $\pm$ 130a	2.69 $\pm$ 0.64a	8.67 $\pm$ 2.58a
SP3	0.10 $\pm$ 0.01a	0.30 $\pm$ 0.01a	92.8 $\pm$ 15.5b	739 $\pm$ 247a	487 $\pm$ 167a	2.34 $\pm$ 0.35a	9.67 $\pm$ 2.42a
Mean	0.11 $\pm$ 0.01	0.40 $\pm$ 0.10	104.9 $\pm$ 23.7	652 $\pm$ 216	375 $\pm$ 130	2.44 $\pm$ 0.22	9.11 $\pm$ 0.51

The significance level is 0.05. Significant differences between soil depths are represented by different lower-case.

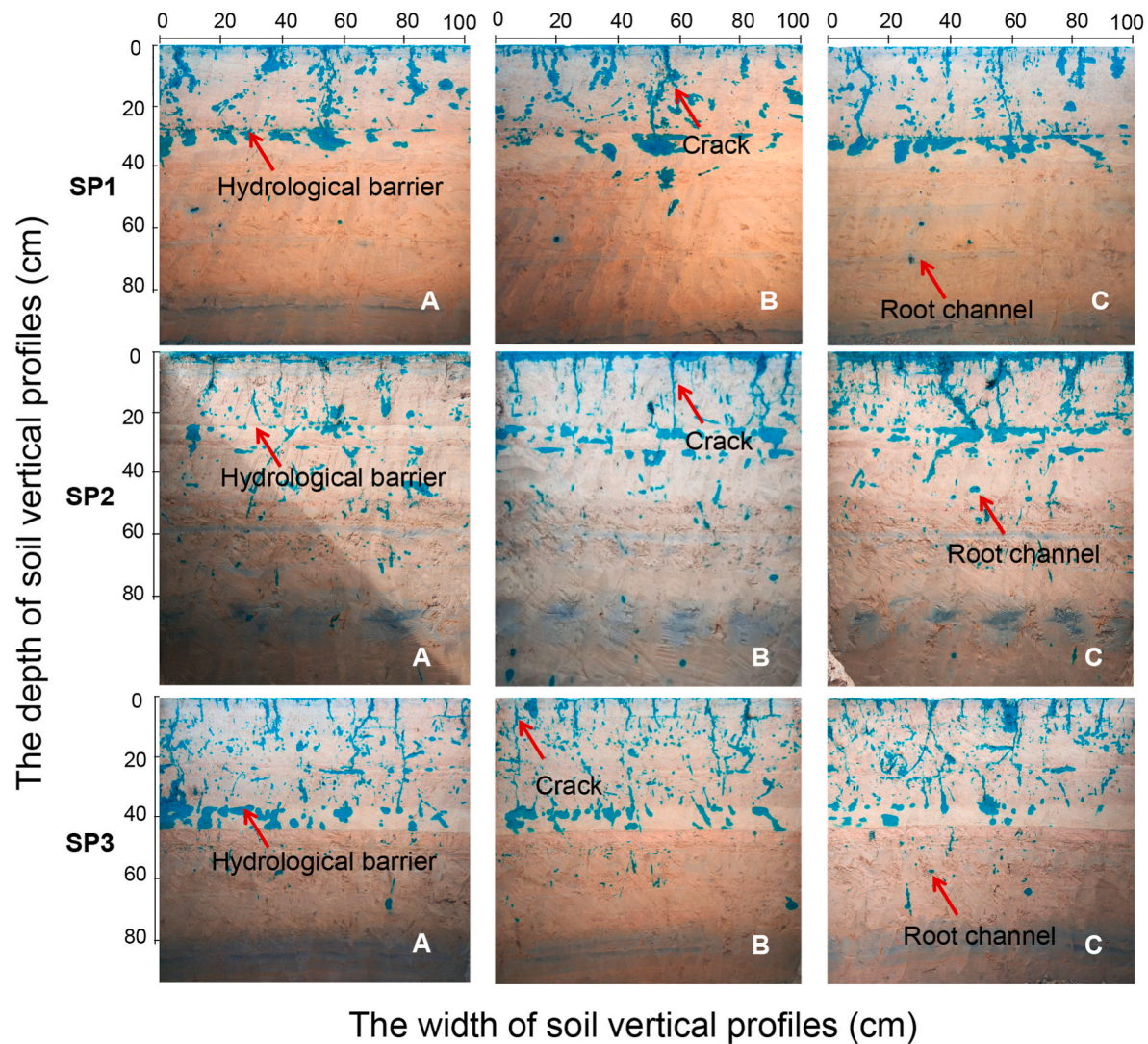


Fig. 2. Dye patterns of vertical soil profiles at sites SP1, SP2, and SP3. A, B, and C referred to vertical soil profiles from the outside of the stained plot to the center. The arrows show the hydrological barrier, cracks, and plant roots of soil.

g. columns 7 and 16), and it decreased rapidly with time before they stabilized. For example, infiltration rate decreased from 13.10 mm/h in the initial infiltration phase to 0.61 mm/h in the steady infiltration phase, with a decrease of 95% in soil column 7 (Fig. 4b). The high value in initial phase was in consistence with previous studies obtained from the Loess Plateau (Ma et al., 2022; Zhang et al., 2019) and from Xishuangbanna Tropical Botanical Garden (Jiang et al., 2019).

Macropore flow by cracks and root channels is a non-equilibrium process which is induced when the solution pressure is close to atmospheric. This non-equilibrium process leads to solution rapidly by-passes

the drier soil matrix and can only be sustained if vertical infiltration rates are large than the lateral infiltration losses into the matrix due to the prevailing pressure potential gradient (Jarvis, 2020). Non-equilibrium water movement and solute transport decreases due to water and solute exchange by diffusion to the matrix (Jarvis, 2020). Besides, water and solute pass through unsaturated soil via macropore without resistance in the initial infiltration phase (Zhang et al., 2017), while they decreased with time since part of these macropores got closed after absorbing water (Qi et al., 2020; Zhang et al., 2019). Hydrophilic minerals, such as illite and montmorillonite, make soils swell after

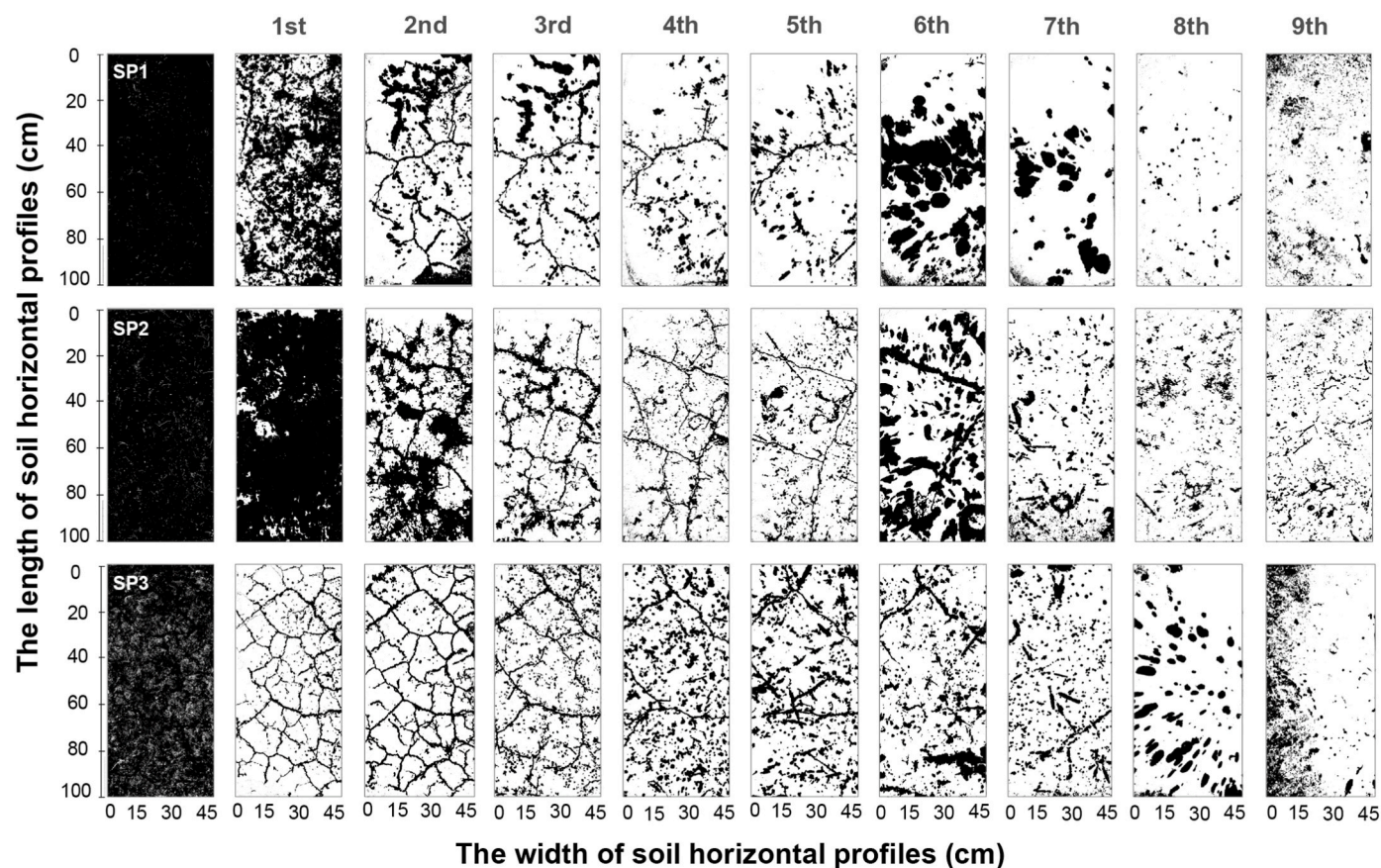


Fig. 3. Binary picture of soil horizontal profiles at different soil depths. 1st, 2nd, 3rd, 4th, 5th, 6th, 7th, 8th, and 9th represent the soil layers at depths of 1, 5, 10, 15, 20, 30, 40, 50, and 60 cm, respectively.

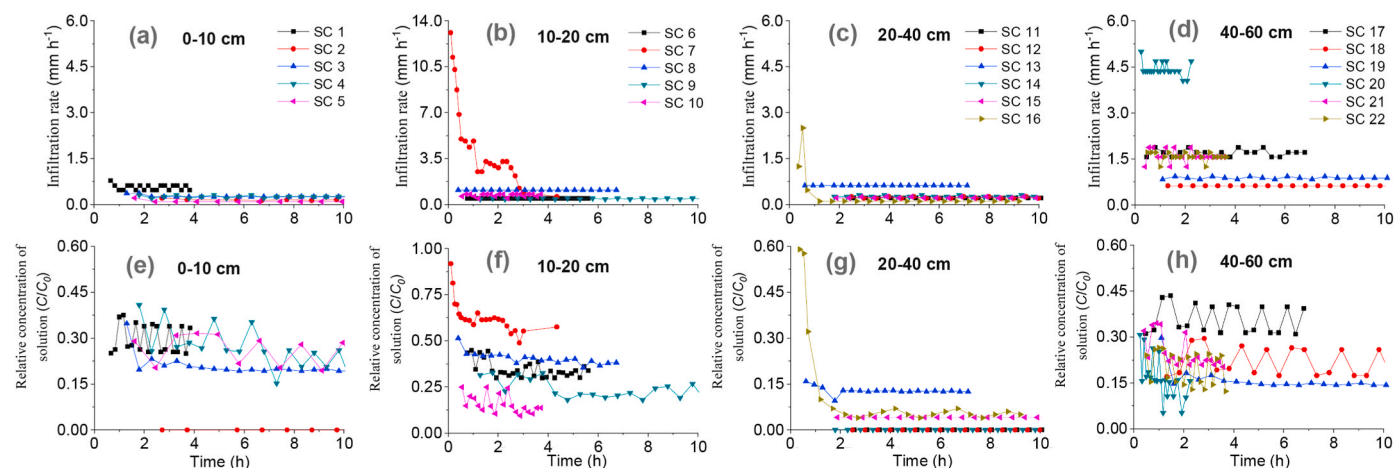


Fig. 4. Curves of infiltration rates and relative concentrations with time for soil columns (SC) 1–5 (a, e), for SC 6–10 (b, f), for SC 11–16 (c, g), and for SC 17–22 (d, h).

absorbing water and shrink after losing water (Ma et al., 2022). We also found that even for soil columns taken from the same soil layer, there are still differences in infiltration rates and concentration (Fig. 4). It is further confirmed macropores such as cracks and root channels could lead to the heterogeneity of water movement and solute transport in soil (Franklin et al., 2021).

Both the initial and stable infiltration rates observed in this study were lower (1.61 mm/h in initial stage and 0.74 mm/h for stable stage) than those obtained in other study areas (Guo et al., 2019; Leung et al., 2015; Zhang et al., 2019). For example, initial and stable infiltration rate

for soil of the *Medicago sativa* grassland in the Loess Plateau were 243.24 and 73.32 mm/h, respectively (Guo et al., 2019). This might be explained by soil properties, as soil permeability increases with the increase of porosity, SOC, and initial water content, whereas decreases with the increase of soil BD (Kang et al., 2022; Wu et al., 2016). High SOC and low BD usually mean more water-conducting macropores, which has an important influence on water holding capacity and water movement (Mossadeghi-Björklund et al., 2016). There was a lack of pores for water movement and solute transport as a result of low SOC and high BD in this study, except cracks and root channels that leading

to the heterogeneity of water and nutrition. Additionally, high initial water content can promote infiltration via enhancing water pressure in the soil (Guo and Lin, 2018b). The relative dry soil in this study could not induce this positive effect on infiltration.

Besides, differences in macropores produced by the root system of different plant may have contrasting effects on the water movement and solute transport (Leung et al., 2015). For example, there were more stable macropores contributing to water infiltration around roots of legumes compared to those around roots of gramineous plants (Wu et al., 2016). In this study, the roots of *S. salsa* were distributed in surface soil (Chen and Sun, 2020; Chen et al., 2016), and only a few were rooted in the deep layer. Although roots of *S. salsa* play an important role in facilitating soil water movement and solute transport, they did not form a dense underground network like other plants.

The differences in infiltration rate and concentration between soil layers were also observed. This might cause by SOC, hydrological barrier, and heterogeneity of water flow paths. We did find a significant higher SOC in surface soil (0–10 cm) than other layers (Table 1), and the more cracks above the hydrological barrier and more root channels below this layer, as well as the concentrated distribution of hydrological barrier at 20–40 cm of soil. SOC closely related to soil texture and structure, such as pore volume, size distribution, and connectivity (Guo and Lin, 2018a; Mossadeghi-Björklund et al., 2016), and has an important impact on water movement and solute transport (Mossadeghi-Björklund et al., 2016). Hydrological barrier usually characterized by higher BD and viscosity (Chelli et al., 2018). This implied that it took more time for the solution to pass through this layer, and more solutes were retained in the soil within short retention time.

3.4. Influence of static connectivity on dynamic connectivity

Static connectivity revealed the structure of vertical connectivity while the dynamic connectivity showed the processes which is also considered as the function of vertical connectivity. The relation between the connectivity structure and function is shown in Fig. 5, static connectivity influenced water movement more significantly while it has

little affection on solute transport. Infiltration rate decreased as dye coverage, fractal dimension, and the number of water pathway logistically increases. However, the concentration changes did not show significant trend as static connectivity indices changes statistically.

Lower dye coverage means water infiltrated mainly through macropores in the soil profile (Flury et al., 1994a; Nimmo, 2021) and in such circumstances, water moves downward quicker than through the whole soil body, with less solutes losses caused by sorption to pore walls and/or diffusion into soil matrix. Generally, the more water pathways, the more it can facilitate water movement in the soil. However, the contrasting result has been found in this study. Water flow is influenced not only by the number of paths but also by pore size (Dai et al., 2020; Jarvis, 2020). It is likely to promote the fast infiltration when there are more macropores in the soil, otherwise, it will retard the flow velocity and increase the residence time (Dai et al., 2020; Jarvis, 2020). Moreover, the pore continuity and tortuosity also controlled the water flow (Beven, 1981; Jarvis, 2020). Fractal dimension gave the information morphological features of the flow paths (Dai et al., 2020; Ogawa et al., 1999; Zhang et al., 2021b). High fractal dimension indicated the soil body have thin and winding paths with stronger lateral water exchange, which is not conductively for fast infiltration.

Besides, the macropores mainly composed by cracks and root channels did promote dynamic connectivity from the water movement perspective. However, concentration changes slightly decreased rather than increased as fractal dimension and water pathway number increase. Solute transport is influenced by multiple factors such as pore binding force and pore water velocity which are many determined by the soil properties and the macropore structures (Jarvis, 2020; Vervoort et al., 1999). Thus, theoretically, static hydrological connectivity may also have a significant influence on the solute transport processes at least statistically. While our experiment gave an opposite result, indicating that solute transport should be more sensitive to the hydraulic condition and concentration gradient in the soil body.

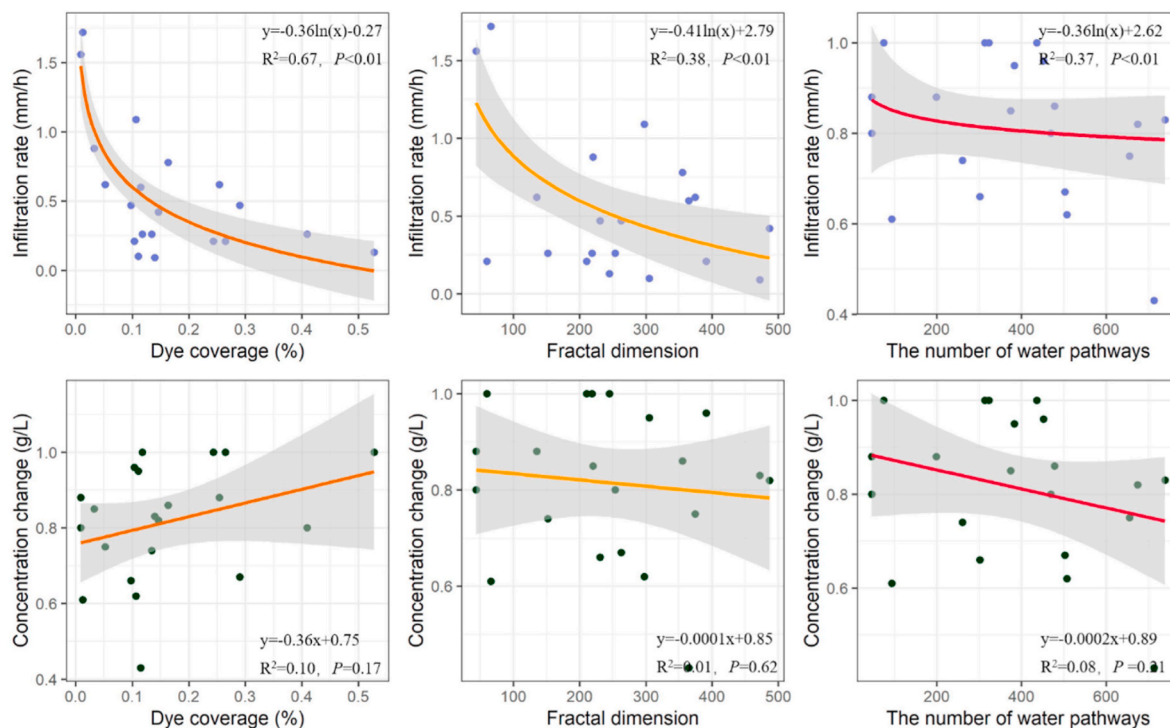


Fig. 5. Relations between static and dynamic hydrological connectivity. Concentration changes (g L^{-1}) refers to the difference between the measured concentration (C) and the applied concentration (C_0).

4. Conclusions

Dye tracer and solute penetration were applied in this study to assess static and dynamic vertical hydrological connectivity, respectively, and the influence of static connectivity on dynamic connectivity were analyzed. We concluded that water flow paths composed of cracks and root channels were irregular and complexity, and thus led to heterogeneous distribution of water and solutes. Non-equilibrium flow characterized by high infiltration rate and concentration in the initial phase was observed in soil with cracks and root channels. The static connectivity had significant effects on water movement while had no effects on solute transport. These findings highlight the importance of cracks and root channels on vertical hydrological connectivity in soil of *S. salsa* community, and suggest linking the static structure of water flow paths with soil water dynamic and heterogeneity to understand soil functions.

Funding

This research was funded by the Fundamental Research Funds for the Central Universities (2022BLRD04).

Data availability statement

Data will be available on reasonable requests to the corresponding author.

CRediT authorship contribution statement

Yuan Cui: Formal analysis, Software, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Ying-Hu Zhang:** Conceptualization, Methodology, Investigation, Resources. **Shi-Jun Zhou:** Investigation. **Yue-Yan Pan:** Investigation. **Rui-Qing Wang:** Investigation. **Zhen Li:** Investigation. **Zhen-Ming Zhang:** Validation, Conceptualization, Writing – review & editing, Supervision, Project administration. **Ming-Xiang Zhang:** Validation, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Abdullah, N., Jain, S., 2020. Multi-index summer flow regime characterization to inform environmental flow contexts: a New England case study. *Ecol. Indic.* 111, 106008.
- Alaoui, A., Caduff, U., Gerke, H.H., et al., 2011. Preferential flow effects on infiltration and runoff in grassland and forest soils. *Vadose Zone J.* 10, 367–377.
- Barbier, E.B., Hacker, S.D., Kennedy, C., et al., 2011. The value of estuarine and coastal ecosystem services. *Ecol. Monogr.* 81, 169–193.
- Bargués Tobella, A., Reese, H., Almaw, A., et al., 2014. The effect of trees on preferential flow and soil infiltrability in an agroforestry parkland in semiarid Burkina Faso. *Water Resour. Res.* 50, 3342–3354.
- Bayraktarov, E., Saunders, M.L., Abdullah, S., et al., 2016. The cost and feasibility of marine coastal restoration. *Ecol. Appl.* 26, 1055–1074.
- Benegas, L., Ilstedt, U., Rounsard, O., et al., 2014. Effects of trees on infiltrability and preferential flow in two contrasting agroecosystems in Central America. *Agric. Ecosyst. Environ.* 183, 185–196.
- Beven, K., 1981. Micro-, Meso-, Macroporosity and channeling flow phenomena in soils. *Soil Sci. Soc. Am. J.* 45, 1245–1245.
- Beven, K., Germann, P., 2013. Macropores and water flow in soils revisited. *Water Resour. Res.* 49, 3071–3092.
- Biehler, A., Chaillou, G., Buffin-Bélanger, T., et al., 2020. Hydrological connectivity in the aquifer–river continuum: impact of river stages on the geochemistry of groundwater floodplains. *J. Hydrol.* 590, 125379.
- Bishop-Taylor, R., Tulbure, M.G., Broich, M., 2018. Impact of hydroclimatic variability on regional-scale landscape connectivity across a dynamic dryland region. *Ecol. Indic.* 94, 142–150.
- Bordoloi, S., Ni, J., Ng, C.W.W., 2020. Soil desiccation cracking and its characterization in vegetated soil: a perspective review. *Sci. Total Environ.* 729, 138760.
- Bracken, L.J., Wainwright, J., Ali, G.A., et al., 2013. Concepts of hydrological connectivity: research approaches, pathways and future agendas. *Earth Sci. Rev.* 119, 17–34.
- Brown, C.J., Adame, M.F., Buelow, C.A., et al., 2021. Opportunities for improving recognition of coastal wetlands in global ecosystem assessment frameworks. *Ecol. Indic.* 126, 107694.
- Burch, G.J., Moore, I.D., Burns, J., 1989. Soil hydrophobic effects on infiltration and catchment runoff. *Hydrol. Process.* 3, 211–222.
- Carless, D., Kulesa, B., Booth, A.D., et al., 2021. An integrated geophysical and GIS based approach improves estimation of peatland carbon stocks. *Geoderma* 402, 115176.
- Chelli, A., Zanini, A., Petrella, E., et al., 2018. A multidisciplinary procedure to evaluate and optimize the efficacy of hydraulic barriers in contaminated sites: a case study in Northern Italy. *Environ. Earth Sci.* 77, 246.
- Chen, B., Sun, Z., 2020. Effects of nitrogen enrichment on variations of sulfur in plant-soil system of Suaeda salsa in coastal marsh of the Yellow River estuary, China. *Ecol. Indic.* 109, 105797.
- Chen, T., Yuan, F., Song, J., et al., 2016. Nitric oxide participates in waterlogging tolerance through enhanced adventitious root formation in the ehalophyte Suaeda salsa. *Funct. Plant Biol.* 43, 244–253.
- Costanza, R., de Groot, R., Sutton, P., et al., 2014. Changes in the global value of ecosystem services. *Global Environ. Change* 26, 152–158.
- Covino, T., 2017. Hydrologic connectivity as a framework for understanding biogeochemical flux through watersheds and along fluvial networks. *Geomorphology* 277, 133–144.
- Cui, Y., Xiao, R., Zhang, M., et al., 2020. Hydrological connectivity dynamics and conservation priorities for surface-water patches in the Yellow River delta National nature Reserve, China. *Ecohydrol. Hydrobiol.* 20.
- Dai, L., Zhang, Y., Liu, Y., et al., 2020. Assessing hydrological connectivity of wetlands by dye-tracing experiment. *Ecol. Indic.* 119, 106840.
- Du, H., Alexander, L.V., Donat, M.G., et al., 2019. Precipitation from persistent extremes is increasing in most regions and globally. *Geophys. Res. Lett.* 46, 6041–6049.
- Feng, L., Xia, J.B., Liu, J.T., et al., 2021. Effects of mosaic biological soil crusts on vascular plant establishment in a coastal saline land of the Yellow River Delta, China. *J. Plant Ecol.* 14, 781–792.
- Fernández, C., Fernández-Alonso, J.M., Vega, J.A., 2020. Exploring the effect of hydrological connectivity and soil burn severity on sediment yield after wildfire and mulching. *Land Degrad. Dev.* 31, 1611–1621.
- Flury, M., Flühler, H., Jury, W.A., et al., 1994a. Susceptibility of soils to preferential flow of water: a field study. *Water Resour. Res.* 30, 1945–1954.
- Flury, M., Flühler, H., Jury, W.A., et al., 1994b. Susceptibility of soils to preferential flow of water: a field study. *Water Resour. Res.* 30.
- Franklin, S.M., Kravchenko, A.N., Vargas, R., et al., 2021. The unexplored role of preferential flow in soil carbon dynamics. *Soil Biol. Biochem.* 161, 108398.
- Freeman, M.C., Pringle, C.M., Jackson, C.R., 2007. Hydrologic connectivity and the contribution of stream headwaters to ecological integrity at regional Scales1. *J. Am. Water Resour. Assoc.* 43, 5–14.
- Granados, V., Gutiérrez-Cánovas, C., Arias-Real, R., et al., 2020. The interruption of longitudinal hydrological connectivity causes delayed responses in dissolved organic matter. *Sci. Total Environ.* 713, 136619.
- Guo, L., Chen, J., Lin, H., 2014. Subsurface lateral preferential flow network revealed by time-lapse ground-penetrating radar in a hillslope. *Water Resour. Res.* 50, 9127–9147.
- Guo, L., Lin, H., 2018a. Addressing Two Bottlenecks to Advance the Understanding of Preferential Flow in Soils. *Adv. Agron* 147, 61–117.
- Guo, L., Lin, H., 2018b. Chapter two - addressing two bottlenecks to advance the understanding of preferential flow in soils. In: Sparks, D.L. (Ed.), *Advances in Agronomy*, vol. 147, pp. 61–117. Academic Press.
- Guo, L., Liu, Y., Wu, G., et al., 2019. Preferential water flow: influence of alfalfa (*Medicago sativa* L.) decayed root channels on soil water infiltration. *J. Hydrol.* 578, 124019.
- Hiatt, M., Passalacqua, P., 2015. Hydrological connectivity in river deltas: the first-order importance of channel-island exchange. *Water Resour. Res.* 51, 2264–2282.
- Jarvis, N.J., 2020. A review of non-equilibrium water flow and solute transport in soil macropores: principles, controlling factors and consequences for water quality. *Eur. J. Soil Sci.* 71, 279–302.
- Jia, X., Shao, M., Yu, D., et al., 2019. Spatial variations in soil-water carrying capacity of three typical revegetation species on the Loess Plateau, China. *Agric. Ecosyst. Environ.* 273, 25–35.
- Jiang, X.J., Chen, C., Zhu, X., et al., 2019. Use of dye infiltration experiments and HYDRUS-3D to interpret preferential flow in soil in a rubber-based agroforestry systems in Xishuangbanna, China. *Catena* 178, 120–131.
- Kang, W., Tian, J., Lai, Y., et al., 2022. Occurrence and controls of preferential flow in the upper stream of the Heihe River Basin, Northwest China. *J. Hydrol.* 607.
- Katuwal, S., Norgaard, T., Moldrup, P., et al., 2015. Linking air and water transport in intact soils to macropore characteristics inferred from X-ray computed tomography. *Geoderma* 237–238, 9–20.
- Leung, A.K., Garg, A., Coo, J.L., et al., 2015. Effects of the roots of *Cynodon dactylon* and *Schefflera heptaphylla* on water infiltration rate and soil hydraulic conductivity. *Hydrol. Process.* 29.
- Li, Y., Zhang, H., Fu, C., et al., 2019. A red clay layer in soils of the Yellow River Delta: occurrence, properties and implications for elemental budgets and biogeochemical cycles. *Catena* 172, 469–479.
- Li, Y., Zhang, H., Tu, C., et al., 2018. Magnetic characterization of distinct soil layers and its implications for environmental changes in the coastal soils from the Yellow River Delta. *Catena* 162, 245–254.

- Liu, J., Engel, B.A., Dai, L., et al., 2019a. Capturing hydrological connectivity structure of wetlands with indices based on graph theory: a case study in Yellow River Delta. *J. Clean. Prod.* 239, 118059.
- Liu, J., Engel, B.A., Zhang, G., et al., 2020. Hydrological connectivity: one of the driving factors of plant communities in the Yellow River Delta. *Ecol. Indic.* 112, 106150.
- Liu, Q., Liu, G., Huang, C., et al., 2019b. Soil physicochemical properties associated with quasi-circular vegetation patches in the Yellow River Delta, China. *Geoderma* 337, 202–214.
- Liu, Z., Cui, B., He, Q., 2016. Shifting paradigms in coastal restoration: six decades' lessons from China. *Sci. Total Environ.* 566–567, 205–214.
- Liu, Z., Fagherazzi, S., Li, J., et al., 2021. Mismatch between watershed effects and local efforts constrains the success of coastal salt marsh vegetation restoration. *J. Clean. Prod.* 292, 126103.
- Ma, J., Zeng, R., Yao, Y., et al., 2022. Characterization and quantitative evaluation of preferential infiltration in loess, based on a soil column field test. *Catena* 213.
- McOwen, C.J., Weatherdon, L.V., Bochove, J.V., et al., 2017. A Global Map of Saltmarshes. *Biodivers Data J.* e11764.
- Morris, C., Mooney, S.J., Young, S.D., 2008. Sorption and desorption characteristics of the dye tracer, Brilliant Blue FCF, in sandy and clay soils. *Geoderma* 146, 434–438.
- Mossadeghi-Björklund, M., Arvidsson, J., Keller, T., et al., 2016. Effects of subsoil compaction on hydraulic properties and preferential flow in a Swedish clay soil. *Soil Tillage Res.* 156, 91–98.
- Murray, N.J., Phinn, S.R., DeWitt, M., et al., 2019. The global distribution and trajectory of tidal flats. *Nature* 565, 222–225.
- Nelson, D.W., Sommers, L.E., 1983. Total carbon, organic carbon, and organic matter. In: Page, A. (Ed.), *Methods of Soil Analysis*.
- Nimmo, J.R., 2021. The processes of preferential flow in the unsaturated zone. *Soil Sci. Soc. Am. J.* 85, 1–27.
- Ogawa, S., Baveye, P., Boast, C.W., et al., 1999. Surface fractal characteristics of preferential flow patterns in field soils: evaluation and effect of image processing. *Geoderma* 88, 109–136.
- Pringle, C.M., 2001. Hydrologic connectivity and the management of biological reserves: a global perspective. *Ecol. Appl.* 11, 981–998.
- Qi, W., Zhang, Z., Wang, C., et al., 2020. Crack closure and flow regimes in cracked clay loam subjected to different irrigation methods. *Geoderma* 358, 113978.
- Sheng, F., Liu, H., Wang, K., et al., 2014. Investigation into preferential flow in natural unsaturated soils with field multiple-tracer infiltration experiments and the active region model. *J. Hydrol.* 508, 137–146.
- Stone, E.L., Kalisz, P.J., 1991. On the maximum extent of tree roots. *For. Ecol. Manag.* 46, 59–102.
- Tan, Z., Li, Y., Zhang, Q., et al., 2021. Assessing effective hydrological connectivity for floodplains with a framework integrating habitat suitability and sediment suspension behavior. *Water Res.* 201, 117253.
- van Schaik, N.L.M.B., 2009. Spatial variability of infiltration patterns related to site characteristics in a semi-arid watershed. *Catena* 78, 36–47.
- Vervoort, R.W., Radcliffe, D.E., West, L.T., 1999. Soil structure development and preferential solute flow. *Water Resour. Res.* 35, 913–928.
- Wang, H., Chen, Y., Chen, Z., 2013a. Spatial distribution and temporal trends of mean precipitation and extremes in the arid region, northwest of China, during 1960–2010. *Hydrol. Process.* 27, 1807–1818.
- Wang, H., Gao, J., Ren, L.-L., et al., 2013b. Assessment of the red-crowned crane habitat in the Yellow River delta nature Reserve, East China. *Reg. Environ. Change* 13, 115–123.
- Wang, L., Zhang, Y., Jia, J., et al., 2021. Effect of vegetation on the flow pathways of steep hillslopes: overland flow plot-scale experiments and their implications. *Catena* 204, 105438.
- Wang, M., Wang, C., Li, Y., 2017. Petroleum hydrocarbons in a water-sediment system from Yellow River estuary and adjacent coastal area, China: distribution pattern, risk assessment and sources. *Mar. Pollut. Bull.* 122, 139–148.
- Ward, J.V., 1989. The four-dimensional nature of lotic ecosystems. *J. North Am. Benthol. Soc.* 8, 2–8.
- Wohl, E., Beckman, N.D., 2014. Leaky rivers: implications of the loss of longitudinal fluvial disconnection in headwater streams. *Geomorphology* 205, 27–35.
- Wu, G., Yang, Z., Cui, Z., et al., 2016. Mixed artificial grasslands with more roots improved mine soil infiltration capacity. *J. Hydrol.* 535, 54–60.
- Wu, Y., Zhang, Y., Xie, L., et al., 2021. Preferential flow improves root-soil system on a small scale: a case study of two ecotypes of *Phragmites communis*. *J. Clean. Prod.* 328, 129581.
- Xia, J., Ren, R., Chen, Y., et al., 2020. Multifractal characteristics of soil particle distribution under different vegetation types in the Yellow River Delta chenier of China. *Geoderma* 368, 114311.
- Xiao, K., Wilson, A.M., Li, H., et al., 2019. Crab burrows as preferential flow conduits for groundwater flow and transport in salt marshes: a modeling study. *Adv. Water Resour.* 132, 103408.
- Xu, J.-J., Zhang, H., Tang, C.-S., et al., 2022. Automatic soil crack recognition under uneven illumination condition with the application of artificial intelligence. *Eng. Geol.* 296, 106495.
- Yin, S., Bai, J., Wang, X., et al., 2020. Hydrological connectivity and herbivores control the autochthonous producers of coastal salt marshes. *Mar. Pollut. Bull.* 160, 111638.
- Yu, X., Zhu, W., Wei, J., et al., 2021. Estimation of ecological water supplement for typical bird protection in the Yellow River Delta wetland. *Ecol. Indic.* 127, 107783.
- Zhang, J., Lei, T., Qu, L., et al., 2019. Method to quantitatively partition the temporal preferential flow and matrix infiltration in forest soil. *Geoderma* 347, 150–159.
- Zhang, Y., Huang, C., Zhang, W., et al., 2021a. The concept, approach, and future research of hydrological connectivity and its assessment at multiscales. *Environ. Sci. Pollut. Control Ser.* 28, 52724–52743.
- Zhang, Y., Jiang, J., Zhang, J., et al., 2021b. Effects of roots systems on hydrological connectivity below the soil surface in the Yellow River Delta wetland. *Ecohydrology* 15, e2393.
- Zhang, Y., Niu, J., Yu, X., et al., 2015a. Effects of fine root length density and root biomass on soil preferential flow in forest ecosystems. *For. Syst.* 24.
- Zhang, Y., Niu, J., Zhang, M., et al., 2017. Interaction between plant roots and soil water flow in response to preferential flow paths in Northern China. *Land Degrad. Dev.* 28, 648–663.
- Zhang, Z.B., Peng, X., Zhou, H., et al., 2015b. Characterizing preferential flow in cracked paddy soils using computed tomography and breakthrough curve. *Soil Tillage Res.* 146, 53–65.